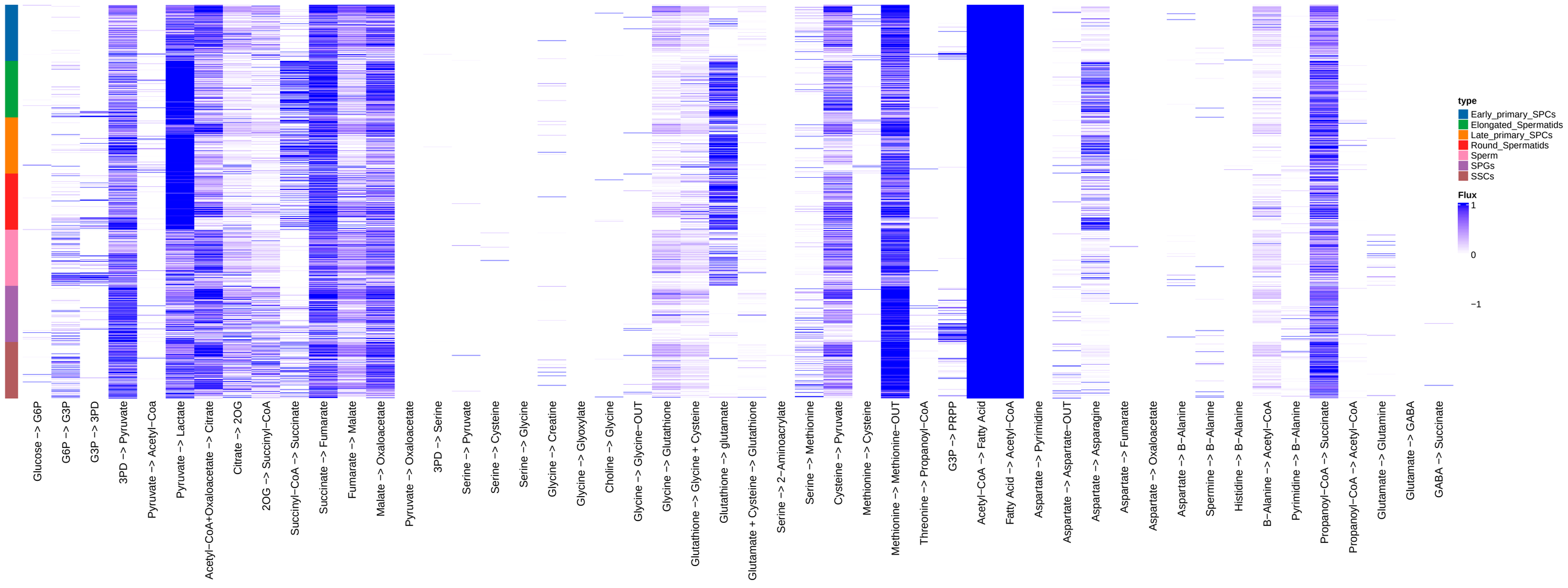
Glucose –> G6P
G6P –> G3P
G3P –> 3PD
3PD –> Pyruvate
Pyruvate –> Acetyl−Coa
Pyruvate –> Lactate
Acetyl−CoA+Oxaloacetate –> Citrate
Citrate –> 2OG
2OG –> Succinyl−CoA
Succinyl−CoA –> Succinate
Succinate –> Fumarate
Fumarate –> Malate
Malate –> Oxaloacetate
Pyruvate –> Oxaloacetate
3PD –> Serine
Serine –> Pyruvate
Serine –> Cysteine
Serine –> Glycine
Glycine –> Creatine
Glycine –> Glyoxylate
Choline –> Glycine
Glycine –> Glycine−OUT
Glycine –> Glutathione
Glutathione –> Glycine + Cysteine
Glutathione –> glutamate
Glutamate + Cysteine –> Glutathione
Serine –> 2−Aminoacrylate
Serine –> Methionine
Cysteine –> Pyruvate
Methionine –> Cysteine
Methionine –> Methionine−OUT
Threonine –> Propanoyl−CoA
G3P –> PRPP
Acetyl−CoA –> Fatty Acid
Fatty Acid –> Acetyl−CoA
Aspartate –> Pyrimidine
Aspartate –> Aspartate−OUT
Aspartate –> Asparagine
Aspartate –> Fumarate
Aspartate –> Oxaloacetate
Aspartate –> B−Alanine
Spermine –> B−Alanine
Histidine –> B−Alanine
B−Alanine –> Acetyl−CoA
Pyrimidine –> B−Alanine
Propanoyl−CoA –> Succinate
Propanoyl−CoA –> Acetyl−CoA
Glutamate –> Glutamine
Glutamate –> GABA
GABA –> Succinate
type
Early_primary_SPCs
Elongated_Spermatids
Late_primary_SPCs
Round_Spermatids
Sperm
SPGs
SSCs
Flux
1
0
−1